

# Free-history timescape: a data-driven void history reconciles the supernova–BAO–CMB geometry but does not resolve the Hubble tension

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Paper I showed that the timescape cosmology’s one-parameter tracker closure cannot jointly fit type Ia supernovae (SNe), baryon acoustic oscillations (BAO), and the cosmic microwave background (CMB) acoustic scale: the SNe pull the present void fraction to  $f_{v0} \simeq 0.85$  while BAO+CMB demand  $\simeq 0.64$  (a  $4.8\text{--}6.6\sigma$  split), though a model-independent free- $E(z)$  check finds the data consistent—the split is the tracker’s rigidity, not a data conflict. Here we free the void history  $f_v(z)$ , keeping the Buchert two-phase average and the wall/void clock-rate dressing. The freed  $f_v(z)$  reconciles Pantheon+, DESI BAO, and the Planck acoustic scale at joint  $\chi^2 = 1396.1$ , below flat  $\Lambda$ CDM (1402.2) and far below the tracker (1469.3), at a plausible  $f_{v0} = 0.640$ ; the amplitude split dissolves to  $0.16\sigma$  (tracker:  $6.5\sigma$ ), though this is a single-amplitude check at fixed jointly-fit shape, not a proof of data consistency. The reconciliation is covariance-dependent: under the cosmology-independent covariance of Lane, Seifert et al. (FLRW-fiducial bias correction removed) the supernova likelihood prefers  $\Lambda$ CDM at every redshift cut ( $\Delta\text{BIC} = +11$  to  $+27$ ). It is moreover kinematic—a backreaction target the Hubble diagram wants, violating Buchert integrability—and fails four tests of independent physical *character* (availability, local-ladder bias, dynamical consistency, early-Universe calibration), facets of one kinematic–dynamical gap: the required history declines by  $\times 3.31$  over  $0 < z < 2.33$  against an observed  $\times 1.16$  (a floor theorem—level matched, shape unavailable); the predicted local bias is  $+2.4\%$  against the  $+8.4\%$  SH0ES requires (pre-registered PARTIAL, failing when the observed catalog is forced in); enforced consistency collapses the three-phase Buchert solve to the tracker, missing the geometry bar by 4953; and the in-model sound horizon  $r_d = 199.6$  Mpc ( $+36\%$ ) drives the bare  $\bar{H}_0$  to  $\sim 32\text{--}40$  km s<sup>−1</sup> Mpc<sup>−1</sup>. The distance geometry reconciles, but the flexibility is kinematic: consistent with Paper I, free-history timescape does not resolve the Hubble tension on present public data.

## I. INTRODUCTION

The proposal that the apparent cosmic acceleration—and, by extension, the Hubble tension—reflects the inhomogeneity of the late Universe rather than a new energy component is the dark-energy-free inhomogeneity programme. The timescape cosmology of Wiltshire [6–8] is its most developed member: the present, void-dominated Universe is modelled as a two-phase Buchert spatial average [9, 10] of spatially-flat “walls” (bound structures, where observers sit) and negatively-curved “voids,” with a quasilocal clock-rate gradient (“dressing”) so that a wall observer fitting Friedmann–Lemaître–Robertson–Walker (FLRW) infers “dressed” parameters distinct from the volume-average “bare” ones. Apparent acceleration follows without dark energy. The sharpest present challenge to the  $\Lambda$ CDM concordance model is the Hubble tension—the  $\sim 5\sigma$  discrepancy between the local distance-ladder value  $H_0 \simeq 73$  km s<sup>−1</sup> Mpc<sup>−1</sup> and the Planck+ $\Lambda$ CDM early-Universe inference  $\simeq 67$  km s<sup>−1</sup> Mpc<sup>−1</sup> [28–30]. Recent Pantheon+ reanalyses with a cosmology-independent covariance report strong Bayesian evidence for timescape over  $\Lambda$ CDM [17, 18], and Wiltshire has since argued that the same void backreaction supplies the cosmological constant itself as an averaging counterterm, requiring no separate dark-energy

source [31]; together these revive the question of whether the inhomogeneity framework also addresses the Hubble tension.

This is the second paper of a three-paper arc. Paper I [1] subjected the standard timescape *tracker*—the one-parameter late-time attractor that fixes the void-fraction history  $f_v(z)$ —to a uniform SN+BAO+CMB harness and found that no single tracker fits the three geometry probes together: the supernovae pull the present void fraction toward  $f_{v0} \simeq 0.85$  while the BAO+CMB geometry demands  $f_{v0} \simeq 0.64$ , a  $4.8\text{--}6.6\sigma$  parameter split. Crucially, a model-independent free- $E(z)$  spline fit to the same three datasets matches  $\Lambda$ CDM: the data are *not* internally contradictory. Paper I therefore diagnosed the tracker’s one-parameter rigidity—not the timescape mechanism, and not a genuine SN-versus-BAO data conflict—as the cause of the combined-fit failure.

The present paper asks the immediate follow-up question: *is the timescape mechanism right but its tracker closure wrong?* We free the void history  $f_v(z)$  from the tracker attractor, retain the Buchert two-phase average and the wall/void clock-rate dressing unchanged, and re-test the framework end to end—first against the SN+BAO+CMB geometry, then against every deeper physical requirement the reconciling history must satisfy.

We state the two-part answer up front, qualitatively; the numbers follow in the body. **Part I** (Sec. III): freeing  $f_v(z)$  *does* reconcile the geometry—the freed history fits Pantheon+, DESI BAO, and the Planck acoustic

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scale below flat  $\Lambda$ CDM and far below the rigid tracker, at a physically plausible present void fraction, and the SN-versus-BAO+CMB amplitude split all but vanishes. Throughout we label this reconciliation *kinematic*: it is a well-defined target the Hubble diagram wants, but the forced history need not satisfy the Buchert integrability conditions and is not, on its face, a general-relativistic solution; it is moreover specific to the supernova covariance adopted. **Part II** (Sec. IV) confronts that history with four requirements of independent physical *character*—is the reconciling history observationally *available*, does it *bias* the local ladder as needed, is it a dynamically *consistent* solution, and is it early-Universe *calibratable*? These are facets of a single kinematic-versus-dynamical gap, not four decorrelated lines of evidence. All four fail: the required steep void-fraction decline is absent from the observed void population (*shape-unavailable*); the predicted local expansion-rate bias falls an order of magnitude short of what SH0ES requires (a pre-registered PARTIAL that fails outright when the observed catalog is forced in); enforcing dynamical consistency collapses the three-phase Buchert solve back to the two-phase tracker (*closed*); and the model’s own sound horizon is far too large, driving the bare Hubble rate well below both canonical values (*fails*). This is *reconciliation without resolution*, and it corroborates and sharpens Paper I’s conclusion.

Every quantitative claim below traces to a committed analysis script and its result file (Sec. VI); each verdict is stated against a pre-registered threshold quoted before the verdict.

## II. FREE-HISTORY TIMESCAPE FRAMEWORK

We use only what is needed; the derivations are in [6, 7, 12] and the uniform harness in [1]. The void volume fraction  $f_v(t)$  grows to a present value  $f_{v0}$  in a two-phase Buchert average of walls and voids [9, 10]. The lapse  $\bar{\gamma} \equiv dt/d\tau_w$  relates volume-average time  $t$  to wall proper time  $\tau_w$ ; in the late-time tracker solution the dressed and bare Hubble parameters obey [7, eq. (B9)]

$$H_0 = g_{\text{dress}}(f_{v0}) \bar{H}_0, \quad g_{\text{dress}}(f_{v0}) = \frac{4f_{v0}^2 + f_{v0} + 4}{2(2 + f_{v0})}, \quad (1)$$

with  $\bar{\gamma}_0 = \frac{1}{2}(2 + f_{v0})$ . The tracker is the *one-parameter* closure that fixes the entire history  $f_v(z)$  from  $f_{v0}$  alone, and has been fitted to multiple probes at a low dressed  $H_0$  [11, 13]. We drop it: we let  $f_v(z)$  be a free function on a redshift grid, solve the dressed geometry (luminosity and transverse-comoving distances, following the DNW13 machinery [12] and the dressed-distance implementation of [16]) for each admissible history, and later confront the freed history with the observed void population.

Freeing the history exposes a lapse-convention choice

that the tracker hides. We carry two readings. *LA (algebraic)* evaluates the lapse from the tracker-form combination  $\frac{1}{2}(2 + f_v)$  node by node. *LB (rate-ratio)* builds the lapse from the local expansion-rate ratio, so it depends on the history’s slope  $f'_v(z)$  and fixes a self-consistent  $\bar{\gamma}_0$  by shooting. LA is the primary reading; LB, together with the no-lapse control (V0), brackets the required history as a lapse-reading systematic used in Part II.

*Solver validation gate.* The general (non-tracker) dressed-geometry solver is a faithful superset of the validated tracker geometry: driven to the tracker history it reproduces the tracker limit to a maximum fractional transverse-distance error  $\lesssim 6.5 \times 10^{-8}$  and an SN  $\chi^2 = 1391.545$  (matching the Paper I tracker value), the LB gate passes at joint tracker  $\chi^2 = 1469.293$  against the reference 1469.293, and it exhibits the genuine non-FLRW signature  $dD_M/dz \neq D_H \equiv c/H(z)$ , which collapses to equality only in the FLRW limit. This is the load-bearing statement that the machinery is correct before we ask what it prefers.

## III. PART I: GEOMETRIC RECONCILIATION

All Part I results are *kinematic*: they establish that the timescape mechanism, with its history freed, has enough shape freedom to fit the distance geometry. Whether that shape is physically realised is the subject of Part II.

### A. Data and harness

We reuse Paper I’s uniform harness and pipeline validation [1]. The supernova data are the Pantheon+ release [19, 20]; the cosmology sample is the  $N = 1580$  SNe with  $z_{\text{HD}} > 0.01$  that are not Cepheid calibrators ( $0.010 < z < 2.26$ ), analysed under the full public statistical+systematic covariance with the absolute-magnitude/ $H_0$  offset marginalised analytically (the supernova verdict has historically been sensitive to the light-curve fitter [14], which the uniform harness holds fixed across models). The main joint fit reuses Paper I’s DR1 SN+BAO+CMB harness for direct comparability with Paper I; DESI DR2 [4, 5] is used as a cross-check (Sec. III C) and throughout Part II. The CMB enters only through the Planck 2018 acoustic-scale point [3]. As in Paper I, the sound-horizon ruler  $r_d$  enters BAO+CMB only through the distance amplitude, so the redshift *shape*—and hence the history  $f_v(z)$ —is independent of the  $r_d$  calibration (a fact we exploit, and whose limits we probe, in Sec. IV D).

### B. The required void history (Probe R)

We fit the freed history jointly to SN+BAO+CMB and compare, on identical data, against the model-independent free- $E(z)$  spline, evolving dark energy

TABLE I. Joint SN+BAO+CMB fit, identical data and harness: minimum  $\chi^2$  of each model. The freed history (headline row) fits below flat  $\Lambda$ CDM and far below the one-parameter tracker; only the model-independent free- $E(z)$  spline does better. The pre-registered reconcile threshold is joint  $\chi^2 \leq 1412.24$ . Column  $k$  is the fitted parameter count; column  $\chi^2 - \chi^2_{\Lambda\text{CDM}}$  is the difference from the  $\Lambda$ CDM row. Raw  $\chi^2$  is compared; no information-criterion preference is claimed. The free-history improvement over  $\Lambda$ CDM carries 4 extra parameters and is not significant (Wilks  $p = 0.19$  on DR1, 0.82 on DR2; BIC favours  $\Lambda$ CDM,  $\Delta\text{BIC} \sim +23$  DR1 /  $+28$  DR2). The Wilks  $p$ -values are approximate— $\Lambda$ CDM is not nested within the freed-history family, so they are indicative rather than exact—and the BIC comparison, which needs no nesting, is the load-bearing one and points conservatively toward  $\Lambda$ CDM. The comparison establishes kinematic shape-sufficiency, not statistical preference.

| Model                                             | $k$    | joint $\chi^2$ | $\chi^2 - \chi^2_{\Lambda\text{CDM}}$ |
|---------------------------------------------------|--------|----------------|---------------------------------------|
| free $E(z)$ spline (model-indep.)                 | spline | 1391.85        | -10.39                                |
| free-history TS ( $f_v(z)$ ) [ <i>this work</i> ] | 5      | 1396.06        | -6.18                                 |
| $w_0w_a$ CDM                                      | 3      | 1398.29        | -3.95                                 |
| flat $\Lambda$ CDM                                | 1      | 1402.24        | 0                                     |
| timescape tracker (1-param)                       | 1      | 1469.29        | +67.06                                |

A no-lapse control (V0) gives  $\chi^2 = 1396.49$ , confirming the result is not an artifact of the lapse convention. Under the cosmology-independent Pantheon+ covariance of Ref. [18] (FLRW-fiducial bias correction removed) the shape-sufficiency fails: the fixed free-history exceeds best-fit  $\Lambda$ CDM by  $\Delta(-2\ln\mathcal{L}) = +18/+24/+35$  at  $z_{\min} = 0.055/0.033/0.010$  (all above the +10 threshold),  $\Delta\text{BIC}(\text{FH} - \Lambda\text{CDM}) = +11.1/+17.0/+27.2$  (fixed-history  $k_{\text{cosmo}} = 0$ ;  $+45.8/+52.3/+64.0$  under an SN-fitted-nodes upper bound); the tracker reproduces Ref. [18]’s  $\Delta\text{BIC}(\text{TS} - \Lambda\text{CDM}) = +4.9/+1.5/-0.5$  (`seifert_covariance_row.json`).

( $w_0w_a$ CDM, the DESI-hinted thawing direction [4]), flat  $\Lambda$ CDM, and the one-parameter tracker (Table I). The pre-registered thresholds [1] are: *reconciles* if joint  $\chi^2 \leq 1412.24$  ( $= \chi^2_{\Lambda\text{CDM}}(\text{DR1}) + 10$ ); *disfavoured* if  $\leq 1427.24$ ; *amplitude-dead* if  $\geq 1469.29$  (the tracker value). Part I is referenced to the DR1  $\Lambda$ CDM baseline ( $\chi^2_{\Lambda\text{CDM}} = 1402.24$ ); Part II’s model-comparison bars are referenced to the DR2 baseline ( $\chi^2_{\Lambda\text{CDM}} = 1399.81$ ). The  $w_0w_a$ CDM comparison is pertinent because inhomogeneity can itself masquerade as evolving dark energy: Ginat and Ferreira [32] show that clumped matter surrounded by voids, small-scale backreaction, and large-scale curvature each distort luminosity and angular-diameter distances into an apparent evolving- $w$  signal. That is a generic light-propagation effect; the freed  $f_v(z)$  tested here is a specific two-phase Buchert realisation of it, with a definite present void fraction and clock-rate dressing, which we hold in Part II to the further dynamical and observational requirements a real backreaction history must meet.

The freed history reaches joint  $\chi^2 = 1396.06$ , comfortably inside the reconcile band and 6.18 below flat  $\Lambda$ CDM; only the model-independent free- $E(z)$  spline

TABLE II. Required void-fraction history  $f_v^{\text{req}}(z)$  at the fit nodes, for the three lapse readings, from the joint SN+BAO+CMB fit. The last column is the LA  $\Delta\chi^2 \leq 1$  band. LA is the primary reading.

| $z$  | LA    | LB    | V0    | LA $\Delta\chi^2 \leq 1$ |
|------|-------|-------|-------|--------------------------|
| 0    | 0.640 | 0.588 | 0.383 | [0.638, 0.641]           |
| 0.3  | 0.531 | 0.497 | 0.250 | [0.521, 0.533]           |
| 0.7  | 0.396 | 0.385 | 0.128 | [0.384, 0.407]           |
| 1.3  | 0.279 | 0.250 | 0.072 | [0.231, 0.318]           |
| 2.33 | 0.194 | 0.121 | 0.017 | [0.148, 0.231]           |

(1391.85) does better, and the rigid tracker (1469.29) is 67 worse. The preferred present void fraction is  $f_{v0} = 0.640$  (LA), a physically plausible value, coincident with the BAO+CMB-preferred value of Paper I and with timescape’s own Planck fits [13]. The associated Hubble rates are a dressed  $H_0 = 63.97 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $g_{\text{dress}}$  convention), a full-rate dressed  $67.65 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , and a global bare  $\bar{H}_0 = 53.79 \text{ km s}^{-1} \text{ Mpc}^{-1}$ . The pre-registered **Verdict: RECONCILES** (mechanism flexible). On DESI DR2 the free-history fit is  $\chi^2 = 1398.28$  versus  $\Lambda$ CDM 1399.81 ( $\Delta = -1.53$ , Wilks  $p = 0.82$ ), still inside the reconcile band; under BIC,  $\Lambda$ CDM is preferred in both DR1 and DR2—consistent with reading the reconciliation as kinematic sufficiency, not model preference.

The reconciliation is robust to the lapse convention. The LB (rate-ratio) sibling fits at  $\chi^2 = 1406.92$  ( $f_{v0} = 0.588$ ;  $+4.68$  above  $\Lambda$ CDM, still inside the reconcile band), and its four validation gates all pass. The required-history nodes for all three lapse readings are collected in Table II: LA is the primary reading; LB and V0 bracket the systematic.

### C. The amplitude split dissolves

Paper I’s headline pathology was that the SN and BAO+CMB data, fitted through the rigid tracker, demand different present void-fraction *amplitudes*. We repeat that split test with the history freed, on DESI DR2 (a free-history joint refit fits at  $\chi^2 = 1398.28$ ,  $f_{v0} = 0.636$ , versus  $\Lambda$ CDM 1399.81 and the tracker 1514.10). Fitting a single amplitude  $A$  to the SN sector and to the BAO+CMB sector separately, and jointly, gives  $A_{\text{SN}} = 0.6351$ ,  $A_{\text{BAO+CMB}} = 0.6363$ ,  $A_{\text{joint}} = 0.6360$ , with a joint excess  $\Delta\chi^2 = 0.027$ —an amplitude split of **0.16 $\sigma$** . The same statistic applied to the tracker as a sanity check is consistent with Paper I’s split:  $f_{v0}^{\text{SN}} = 0.853$  against  $f_{v0}^{\text{BAO+CMB}} = 0.636$ ,  $\Delta\chi^2 = 42.85$ , a **6.5 $\sigma$**  split—a single-amplitude  $\Delta\chi^2$  statistic recovering the same 0.85-versus-0.64 tension Paper I established with a profile-likelihood parameter shift. The split *dissolves* once the history shape is free.

The interpretation is exactly Paper I’s diagnosis, now

demonstrated constructively: the SN and BAO+CMB data “want” the *same* amplitude once the history is free to bend. The 0.85-versus-0.64 split was the tracker parametrisation’s rigidity, not a data conflict. One caveat bounds the claim: the free-history split constrains only the single overall amplitude  $A$  with the jointly-fit shape held fixed (the four extra shape degrees of freedom having already absorbed the SN-versus-BAO+CMB tension), so the  $0.16\sigma$  demonstrates that the tracker’s rigidity is removable, not that the data are intrinsically consistent under this model.

#### D. Why this is only a kinematic reconciliation

The reconciliation of Secs. III B–III C is a statement about distance geometry, not about dynamics. The forced  $f_v(z)$  *violates the Buchert integrability condition*. For a genuine single-phase two-scale GR solution the void curvature parameter  $\alpha^2 = -k_v f_{vi}^{2/3}$  must be constant in bare time (DNW13 Eq. 11 [12]); on the tracker it is constant to the  $\sim 2 \times 10^{-6}$  pipeline floor, whereas the forced required history drifts by  $\sim 152\%$  across its interior (Sec. IV C). The freed history is therefore “the backreaction the Hubble diagram wants”—a well-defined kinematic *target*, not a demonstrated general-relativistic solution. The geometric reconciliation is moreover specific to the supernova covariance: repeating the SN comparison under the cosmology-independent covariance of Refs. [17, 18]—which removes the FLRW-fiducial bias correction the timescape programme regards as circular—the fixed free-history is disfavoured relative to  $\Lambda$ CDM at every redshift cut (Table I, footnote), so the shape-sufficiency of Sec. III B is not robust to that choice. (The evaluation is deliberately fixed-shape and conservatively counted: the nodes are held at their Part I values and charged zero SN-side parameters, so the free-history enters with *fewer* parameters than  $\Lambda$ CDM and the disfavouring is a likelihood gap,  $-2\Delta \ln \mathcal{L} = +18$  to  $+35$ , not a parameter penalty. Refitting the nodes under the covariance of Refs. [17, 18] would define a *different* kinematic target, whose Part II tests would then require their own run; the question posed here is whether the Part I reconciling history itself is covariance-robust, and it is not.) Part II asks, at full detail, whether that target is available in the sky, consistent as dynamics, and calibratable to the early Universe.

### IV. PART II: DOES THE RECONCILIATION SURVIVE?

We report the four Part II tests at the same length and specificity as the Part I reconciliation. Each states its pre-registered threshold before its verdict. All four fail; the honest negative result is the value of the paper. Every headline number in both parts was reproduced by an independent adversarial re-derivation against its pre-

registered threshold, with the per-check records committed alongside the code (Acknowledgments and Sec. VI).

#### A. Availability: is the required history in the sky?

The reconciling history is a *prediction* about the void population: the required  $f_v(z)$  must decline steeply with redshift. Over  $0 < z < 2.33$  the LA required history declines by a total factor  $\times 3.31$  (Table III). We confront this with the *observed* void volume fraction, measured from the BOSS DR12 galaxy sample with a watershed/ZOBOV void finder (Mao et al. 2017 [21]), calibrated to the  $z = 0$  2M++ field [26] and cross-checked against the watershed filling-fraction literature [22, 25]. The measurement chain is insensitive to the standard void observational systematics: the  $z = 0$  anchor is the real-space-reconstructed 2M++ field; the  $z > 0$  route uses the growth *ratio*, in which the redshift-space-distortion boost cancels to first order; the light-cone gradient across an individual void (its far wall seen  $\sim 2 \times 10^8$  yr earlier) is  $< 0.2\%$  in the volume fraction; and per-sightline propagation effects are bounded by Paper I’s line-of-sight nulls ( $\lesssim 0.05$  mag RMS).

The observed void fraction declines by a total factor of only  $\times 1.16$  over the same range—the population is far flatter than the required history. At the pivotal node  $z = 0.7$  the observed  $f_v^{\text{obs}} = 0.601$  against a required 0.396, a gap of 0.205 ( $\sim 52\%$  relative). This gap is not a statistical undershoot but a *floor theorem*: the below-mean void fraction—the volume fraction with density below the cosmic mean—is at least half the volume ( $\geq 0.5$ ). This holds because the late-time volume-weighted density PDF is right-skewed: for a lognormal field  $P(\delta < \bar{\delta}) = \Phi(\sigma/2) \geq 1/2$ , reaching  $1/2$  only in the Gaussian (early-time) limit, and gravitational collapse drives the skewness that way. Empirically the fraction is  $\geq 0.56$  across  $R_s \in \{2, 4, 8\} h^{-1} \text{Mpc}$ , giving the  $z = 0.7$  band  $[0.560, 0.654]$ . The required  $f_v(0.7) = 0.396$  lies *below* the 0.5 floor, so no admissible below-mean population can reach it: the gap is unbridgeable by the definition, not a large-significance undershoot. The pre-registered SHAPE test declares the history *supplied* only if the observed decline ratio lies inside the required band at every node; here the observed ratio exceeds the required-hi edge at every node above zero. The  $z = 0$  *level* matches (observed 0.643 inside  $[0.614, 0.672]$ ; required  $f_{v0} = 0.640$  passes), so it is specifically the *shape*, not the amplitude, that is unavailable.

**Verdict: SHAPE-UNAVAILABLE.** A fit-space corroboration sharpens this: feeding the survey-measured history in with *zero* free parameters ( $f_v(z) = f_v^{\text{obs}}(z)$  fixed) gives a forced joint  $\chi^2 = 2845.6$  on DR2, missing the (one-fewer-parameter) BIC bar [27] of 1407.2 by 1438; the miss is  $> 1400$  under every covariance treatment tested (`verify_forced_joint_fit.json`). The theoretical two-phase  $f_v$  is a volume-average quantity that need not map onto any single void-finder definition;



TABLE III. Required versus observed void-fraction decline.  $f_v^{\text{req}}$  is the LA required history (Table II);  $r_{\text{req}}$  is its pre-registered decline-ratio band  $f_v(z)/f_v(0)$  (mapping forecast);  $f_v^{\text{obs}}$  and  $r_{\text{obs}}$  are the BOSS DR12 below-mean measurement at the required grid. The  $z = 0$  level matches (observed 0.643 inside the required band [0.614, 0.672]); it is the *shape* (decline) that is unavailable. The observed decline is flatter than the required band at every node above zero. Total decline  $z = 0 \rightarrow 2.33$ : required  $\times 3.31$ , observed  $\times 1.16$ .

| $z$  | $f_v^{\text{req}}$ (LA) | $r_{\text{req}}$ band | $f_v^{\text{obs}}$ | $r_{\text{obs}}$ | status                                                               |
|------|-------------------------|-----------------------|--------------------|------------------|----------------------------------------------------------------------|
| 0    | 0.640                   | — (anchor)            | 0.643              | — (anchor)       | LEVEL-PASS (band [0.614, 0.672])                                     |
| 0.3  | 0.531                   | [0.812, 0.836]        | 0.623              | 0.968            | FAIL: observed flatter                                               |
| 0.7  | 0.396                   | [0.599, 0.637]        | 0.601              | 0.934            | FAIL: observed flatter (gap 0.205, 52%)                              |
| 1.3  | 0.279                   | [0.361, 0.499]        | 0.578              | 0.898            | FAIL (growth-extrap.): $r_{\text{obs}} > r_{\text{req}}^{\text{hi}}$ |
| 2.33 | 0.194                   | [0.231, 0.363]        | 0.555              | 0.863            | FAIL (growth-extrap.): $r_{\text{obs}} > r_{\text{req}}^{\text{hi}}$ |

Observed  $f_v$  measured on the primary below-mean route at  $z = 0.3$  and  $z = 0.7$  (BOSS DR12,  $R_s = 4 h^{-1} \text{Mpc}$ ); the  $R_s$ -sensitivity band at  $z = 0.7$  is [0.560, 0.654] across  $R_s \in \{2, 4, 8\} h^{-1} \text{Mpc}$ , all  $\geq 0.5$  (floor holds). The  $z = 1.3$  and 2.33 observed entries are  $\Lambda\text{CDM}$ -growth extrapolations of the  $z \leq 0.7$  BOSS DR12 below-mean measurement (propagated with the validated  $\Lambda\text{CDM}$   $D(z)$  growth ratio), not direct measurements; at both nodes the observed ratio exceeds the required-hi edge (pre-registered SHAPE test).

accordingly the verdict is not that one definition falls short but that *no* tested observational definition supplies both the  $z = 0$  level and the decline. The below-mean definition matches the level ( $f_v(0) = 0.643$  in the band [0.614, 0.672]) but is only  $\times 1.16$ -flat, whereas a deep-void definition ( $\delta < -0.3$ ) matches or exceeds the required  $\times 3.31$  decline but starts too low ( $f_v(0) \sim 0.44\text{--}0.48$ , failing the level)—no single void definition occupies both corners. The  $z = 0$  level-pass is moreover specific to the LA reading: LB ( $f_{v0} = 0.588$ ) and V0 (0.383) fall below the below-mean band, so availability is worse under the lapse-reading systematic.

A commensurable and definition-free test is already within reach: the void-fraction redshift evolution can be measured directly in the full-general-relativity numerical-relativity watershed catalogs of Williams et al. [25] already used here for the  $z = 0$  bracket, sidestepping the galaxy-survey definition ambiguity entirely. On the observational side, newer and higher-redshift void catalogs—DESIVAST from the DESI DR1 Bright Galaxy Survey [33] and the eBOSS void sample spanning  $0.6 < z < 2.2$  [34]—can tighten the  $z = 0$  level and extend the observed decline across the full required range; we claim no such recomputation here, only that the measurement is sharpenable.

### B. The local expansion-rate bias $b_{\text{pred}}$

For the model to resolve the Hubble tension *as a local measurement bias*, its own two-scale structure must predict that a wall observer on the second rung of the SH0ES distance ladder measures an apparent  $H_0$  high by a specific amount. Define the target from the model’s internal ladder-versus-global bare Hubble offset,

$$b_{\text{req}} = \frac{\bar{H}_0^{\text{anchored}}}{\bar{H}_0^{\text{global}}} - 1 = +8.4\% \quad (b_{\text{req}} = 0.0842), \quad (2)$$

which is the excess any local-bias mechanism must reproduce (the anchored fit that fixes the numerator

passes an  $\Lambda\text{CDM}$  validation gate at  $H_0 = 73.53 \pm 1.02 \text{ km s}^{-1} \text{Mpc}^{-1}$ , consistent with the SH0ES distance ladder [2], recently re-anchored with HST Small Magellanic Cloud Cepheids [35] and consolidated across estimators by the local distance-ladder consensus [36]). Equation (2) is a *bare*-frame rate ratio ( $58.32/53.79 - 1$ ), and it is the *smallest* of the admissible targets: the corresponding dressed-frame ratio ( $\sim 73.5/64.0 - 1 \approx 0.15$ ) is nearly twice as large, so reading  $b_{\text{req}}$  in the dressed frame would only widen the under-prediction found below. The verdict is therefore frame-robust.

*The apparent- $H_0$  window and the void-scale ceiling.* The tracker-form combination

$$E_{\text{max}}(f_{v0}) = \frac{3/2}{g_{\text{dress}}(f_{v0})} - 1, \quad (3)$$

with  $g_{\text{dress}}$  from Eq. (1), is an algebraic identity that, evaluated at Wiltshire’s two published present void fractions, reproduces the 17–22% window as a consistency check (not an independent derivation):  $E_{\text{max}}(0.76) = 0.171$  and  $E_{\text{max}}(0.695) = 0.220$  (validation gate a, matching the literature window [7, 8, 15]). It is used only for this window check and does *not* give the ceiling adopted below—at the fitted  $f_{v0} = 0.640$  it returns only 0.2615. The void-scale ceiling adopted below is a different quantity, the pure wall-observer clock conversion

$$E_{\text{max}} = \bar{\gamma}_0 H_{v0}/H_{\text{dress},0} - 1 = 0.3365 \quad (\text{LA}; 0.208 \text{ LB}), \quad (4)$$

evaluated on the fitted free history (gate b; this corrects an earlier mislabeling of the volume-average 0.195 as “the maximum”). The “maximum apparent- $H_0$  excess” is thus a modelling choice spanning a factor  $\sim 3$  across conventions (tracker route 0.09–0.12; volume-average 0.195; void-scale maximum 0.337); we adopt the largest, model-favourable value, and the FAILS verdict below is robust to the choice.

*The SH0ES geometry dilutes the ceiling.* The second-rung Hubble-flow sample does not sit at the void-scale

maximum. The survey-averaged bias is

$$b_{\text{pred}}^{\text{survey}} = E_{\text{max}} \langle \varphi \rangle_{\text{HF}} = 0.0240, \quad (5)$$

the void weight averaged over the  $n_{\text{HF}} = 277$  Hubble-flow SNe *alone* ( $\langle \varphi \rangle_{\text{HF}} = 0.0713$ ); the  $n_{\text{calib}} = 77$  calibrators ( $\langle \varphi \rangle_{\text{calib}} = 0.994$ ) do *not* enter, because the SH0ES  $M_B$  anchor is set by geometric Cepheid distances (parallax/maser/detached-eclipsing-binary), unbiased by the local apparent rate. This is a  $14\times$  dilution of the ceiling  $E_{\text{max}} = 0.3365$ . Two bracketing *diagnostics* (neither is the primary): the differential  $E_{\text{max}}(\langle \varphi \rangle_{\text{calib}} - \langle \varphi \rangle_{\text{HF}}) = 0.310$  is physically inapplicable (the calibrators sit deep in the void), and the co-average over both populations,  $0.092$ , has no physical basis. The lensing-*physical* void rate (calibrated to stacked void lensing densities [23, 24]) gives  $b_{\text{pred}}^{\text{phys}} = 0.0158$ , band  $[0.0150, 0.0287]$ . The entire admissible  $\varphi$ -band,  $[0.0004, 0.0720]$ , lies *below*  $b_{\text{req}} = 0.0842$ : a one-sided under-prediction.

*Verdict against the pre-registered threshold.* The P3 threshold (P3 and P4 are Paper I’s pre-registered local-bias thresholds [1], on the bias *amplitude* and on its *z-profile* respectively) is  $|b_{\text{pred}} - b_{\text{req}}| \leq 1\sigma \Rightarrow \text{RESOLVES}$ ;  $\leq 2\sigma \Rightarrow \text{PARTIAL}$ ; else **FAILS**. The decisive statement needs no  $\sigma$ : the entire admissible  $\varphi$ -band  $[0.0004, 0.0720]$  lies below  $b_{\text{req}} = 0.0842$ , so the mechanism under-predicts the required survey-averaged bias for *every* admissible  $\varphi$  (one-sided). Matching  $b_{\text{req}}$  would require a profile-decay scale  $r_{\text{hom}}^* = 140 h^{-1} \text{Mpc}$ , outside Wiltshire’s declared  $[80, 120] h^{-1} \text{Mpc}$  band and beyond the  $\sim 137.5 h^{-1} \text{Mpc}$   $2M++$  homogeneity scale—where the density field is already homogeneous, so no apparent- $H_0$  bump can persist. Against the P3 tree the pre-registered significance is  $n\sigma_{\text{prereg}} = 1.57$  (PARTIAL), softened only by the wide  $\varphi$ -shape *theoretical* systematic, which the one-sided envelope shows *double-counts*  $\sigma_\varphi$ ; the measurement-only  $n\sigma_{\text{meas}} = 4.16$  is the systematic-*excluded* bound, reported for completeness, not the basis of the verdict.

*Catalog-forced cross-check (the symmetry rule).* Substituting the *observed* void population (Sec. IV A) for the joint-fit winner makes the mechanism fail harder, not softer: the anchored ceiling drops to  $E_{\text{max}}^{\text{obs}} = 0.0657$  and  $b_{\text{pred}}^{\text{survey,obs}} = 0.0047$ , while the forced global bare rate craters to  $\bar{H}_0 = 50.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$  (missing the BIC bar by 1438). The under-prediction ratio  $b_{\text{req}}/b_{\text{pred}}$  is  $17.9\times$  against the fitted target and  $68.1\times$  against the (larger) catalog-forced target.

**Verdict: PARTIAL** (pre-registered) / **FAILS** (one-sided envelope and catalog-forced). The pre-registered P3 outcome is PARTIAL, at  $1.57\sigma$  (the most-permissive of the registered readings; all lie below  $2\sigma$ , PARTIAL), and we foreground it as the registered result. The family-level **FAILS** is the stronger reading: it rests on the one-sided envelope—the entire admissible  $\varphi$ -band lies below  $b_{\text{req}}$ —and on the catalog-forced test, in which the observed void population under-predicts  $b_{\text{req}}$  by  $68\times$ ; it does not rest on relabeling the  $4.16\sigma$  measurement-only significance as robust. The envelope is decisive only

because it discards  $b_{\text{req}}$ ’s own uncertainty, exactly the width the pre-registered  $1.57\sigma$  retains, so the two readings bracket the result rather than contradict. Either way the predicted local bias is an order of magnitude below the required  $+8.4\%$ —whether the void structure is fitted to the Hubble diagram or supplied by the telescope—killing the local-bias resolution for this model family regardless of Part I’s joint-fit quality.

### C. Dynamical consistency: is the required history a GR solution?

Section IIID noted that the forced history violates Buchert integrability; here (work package WP-B, the dynamical three-phase test) we quantify it and make the honest attempt to *restore* consistency. The invariant  $\alpha^2 = -k_v f_v^{2/3}$  (DNW13 Eq. 11 [12]) must be constant in bare time for a genuine single-phase two-scale GR solution. Its fractional drift across the interior of the forced required history is  $\sim 152\%$ . The magnitude is representation-dependent—the five-node history under-determines  $f_v''$ , so the drift reads  $\sim 152\%$  on the boundary-condition-robust interior window,  $\sim 1550\%$  over the full span, and  $\sim 2150\%$  under the solver’s monotone-PCHIP interpolation—but every reading exceeds the tracker’s  $2 \times 10^{-6}$  pipeline floor by six-plus orders of magnitude, so the integrability violation is robust to the representation. The observed history drifts by  $\sim 5\times$  (span). An exact-analytic transcription check drifts only at machine precision, so the drift is physical, not numerical. *No consistent single-phase Buchert model tracks either history.*

The honest attempt to make it consistent is a *three-phase* Buchert solve (shallow void, deep void, wall), which adds the dynamical degrees of freedom that a single phase lacks. All four solver gates pass—G1 tracker-limit ( $\text{SN } \chi^2 = 1391.545$ ), G2 integrability, G3 lapse-limit (agreement with  $\frac{1}{2}(2 + f_v)$  to  $4 \times 10^{-16}$ ), G4 numerics—and the stage verdict is GO. But the best fit is *degenerate*: with the four constants pinned by the measured structure (the primary  $k = 0$  configuration), the shallow void phase’s amplitude drives to  $\alpha_{s2} \rightarrow 0$ , i.e. the three-phase model collapses to the two-phase empty-void tracker with a single genuine void of fraction  $f_{d0} \simeq 0.27$ . The tracking solution “tracks” only by becoming the tracker.

The pre-registered geometry bar (DR2) is  $\chi^2 \leq \chi_{\Lambda\text{CDM}}^2(\text{DR2}) + \ln N \approx 1407.2$ . The zero-parameter ( $k = 0$ ) forced-geometry prediction of the best three-phase constants gives a joint  $\chi^2 = 6360.2$ , missing the bar by **4953**; a degeneracy check confirms this equals the tracker prediction at  $f_{v0} = f_{d0} = 0.274$  to  $0.02$  in  $\chi^2$ . Even the never-a-claim *constants-freed* diagnostic ceiling (the T1 ceiling of Table IV)—a different configuration, freeing the four constants the theory pins by structure, in which the *deep*-phase amplitude collapses instead ( $\alpha_{d2} \rightarrow 7 \times 10^{-7}$  at  $f_{d0} = 0.022$ )—misses the bar by 107 ( $\Delta\text{BIC} = +136$  versus  $\Lambda\text{CDM}$ ).

**Verdict: CLOSED** (a theory-level close-out, not a numerical failure). The required history is not a general-relativistic single-phase solution, and the dynamical three-phase attempt to build one collapses back to the tracker it was meant to generalise.

#### D. The sound horizon: does it survive to the early Universe?

Part I’s fit borrowed the external drag-epoch sound horizon  $r_d = 147.09$  Mpc as a ruler. WP-C (the sound-horizon work package) replaces that crutch with the model’s own bare-frame radiation-era physics (DNW13 machinery [12]) and asks what  $r_d$  free-history timescape actually predicts. The answer is

$$r_d^{\text{in-model}} = 199.6 \text{ Mpc}, \quad (6)$$

+35.7% above the external drag-epoch ruler  $r_d = 147.09$  Mpc (the in-model decoupling-epoch radius is  $r_{\text{dec}} = 195.9$  Mpc). The machinery is validated: its  $\Lambda$ CDM gate returns  $r_* = 144.3$  Mpc and  $r_{\text{drag}} = 146.9$  Mpc, matching the expected values.

This in-model  $r_d$  stands in tension with the timescape literature, which we own rather than paper over. Fitting the CMB acoustic peaks within timescape, Nazer and Wiltshire [13] obtained an *internally consistent* acoustic scale at a dressed  $H_0 \approx 61 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ; our bare-frame radiation-era treatment instead returns  $r_d = 199.6$  Mpc and a bare rate far below that fit. The two are not yet reconciled: our machinery is validated only on the  $\Lambda$ CDM gate above, and the disagreement lies in how the two calculations treat bare-frame recombination. Reconciling them is an explicitly open task for the timescape programme—one it already flags as unfinished, Seifert et al. [18] noting that calibration of the primordial sound speed and the BAO scale remains to be done—rather than a settled result of this paper. We therefore report the in-model  $r_d$  as a self-consistency *failure* of the freed history, not as a determination of the timescape sound horizon.

Part I’s distance fit (Sec. III B) is untouched by the in-model  $r_d$ : its  $r_d$ -independence comes from the analytic  $\alpha$ -marginalization of the BAO+CMB amplitude (Sec. III A), which absorbs any overall ruler rescaling—not from the near-unity acoustic ratio. The in-model geometry nonetheless disagrees with CMB early-Universe physics: although the acoustic ratio  $r_*/r_{\text{drag}}$  matches Planck to 0.03% (0.9816 vs 0.9819), the in-model acoustic angle  $\ell_A = 218.9$  fails Planck’s 301.7 by 27% (291.9 vs 301.7 even when rescaled by the external  $r_d$ ), with bare shift  $R = 1.11$  vs 1.74 and a BBN- $\eta$  baryon density  $\omega_b = 0.0186$  vs 0.0224. The Part I CMB point therefore constrains only the redshift *shape*, not agreement with CMB early-Universe physics. But the *absolute* Hubble rate is not  $r_d$ -independent. With the in-model  $r_d$ , the global bare rate drops from  $\bar{H}_0 = 53.8$  to  $39.6 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , and the self-consistent fixed point is

$\bar{H}_0^* = 32.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$  (at  $r_d^* = 244.5$  Mpc), with the corresponding dressed rates falling to  $47.2$  ( $g_{\text{dress}}$ ) and  $49.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$  (full-rate)—far below both the reconciling band and SH0ES. Crucially, the failure is not an artifact of the borrowed ruler: sound-horizon-independent  $H_0$  determinations, which never invoke  $r_d$ —strong-lensing time delays [37], water-megamaser geometric distances ( $73.9 \pm 3.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ) [38], and gravitational-wave standard sirens [39]—all sit in the upper-60s to mid-70s, far above the  $\sim 47\text{--}50 \text{ km s}^{-1} \text{ Mpc}^{-1}$  dressed rates that a self-consistent in-model  $r_d$  implies.

The pre-registered question is whether the Part I *absolute- $H_0$*  claim survives an in-model sound horizon; it does not ( $R1_{\text{survives}} = \text{false}$ , the pre-registered survival flag). As in Paper I, we distinguish resolution from recasting: driving the bare Hubble rate *below* both canonical values is a recast of the tension, not a reconciliation near the local  $H_0$ .

**Verdict: FAILS** (as an absolute- $H_0$  claim). The model’s own early-Universe physics predicts a sound horizon  $\sim 36\%$  larger than the ruler it had borrowed; forcing self-consistency collapses the bare Hubble rate to  $\sim 32\text{--}40 \text{ km s}^{-1} \text{ Mpc}^{-1}$ .

#### E. Self-inflicted tensions (WP-N)

Table IV aggregates the program’s tension ledger (work package WP-N): for each tension, the free-history (kinematic) status, whether it is *self-inflicted* by the kinematic reconciliation, and the source artifact. The tally is 6 YES self-inflicted, 2 partial, and 1 target (the Hubble tension itself, which the model exists to resolve). The synthesis is that the mechanism is CLOSED at every level tested: the void geometry is shape-unavailable, the local SH0ES bias fails, the dynamical three-phase consistency is closed, and the sound-horizon absolute- $H_0$  reconciliation fails.

The growth/ $S_8$  row deserves separate note: it is *undefined-in-kinematic-reading*. The kinematic reconciliation has no perturbed metric, so there is no growth equation on the averaged background— $f\sigma_8$ , RSD  $\beta$ , weak-lensing/ $S_8$ , cluster counts, and the ISW are none of them rigorously defined. The measured  $\sigma_m(L = 20 h^{-1} \text{ Mpc}, z)$  count-in-cells amplitude ( $A = 0.444$ , reduced  $\chi^2 \approx 0.22$ ) is an *input*, not a prediction. This is a structural boundary of the kinematic reading, not a computed defeat, and it is defined only after the (closed) dynamical WP-B reading is built.

Two further ledger rows carry only a status token in Table IV and warrant a sentence each, both having been run against pre-registered thresholds. The *void-wall contrast budget* asks whether real (lensing/2M++-measured) void and wall contents can supply the fitted expansion contrast  $(H_v - H_w)/\langle H \rangle \simeq 0.474$  at  $z = 0$ ; the pre-registered expectation was an available contrast 0.25–0.45 against a required 0.45–0.5—a shortfall expected but not presumed, with the explicit falsifier that a two-sided

TABLE IV. WP-N self-inflicted tensions of the free-history (kinematic) reading. “Self-inflicted?” marks tensions created *by* the kinematic reconciliation rather than inherited. Tally: 6 YES, 2 partial, 1 target. The growth/ $S_8$  row (8) is structurally, not numerically, undefined: with no perturbed metric the kinematic reading cannot even pose the  $S_8$  question.

| Tension                                 | Free-history (kinematic) status                                                                                                                                                                                                                                                                                           | Self-infl.? | Source                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------|
| Hubble (SH0ES vs. early)                | FAILS: $b_{\text{pred}}^{\text{survey}}$ ceiling 0.0240 + physical 0.0158 vs. $b_{\text{req}} = 0.0842$ ; pre-registered $n\sigma_{\text{prereg}} = 1.57$ (PARTIAL), with the systematic- <i>excluded</i> , measurement-only $n\sigma = 4.16/4.73$ (ceiling/physical) reported for completeness, not as the verdict basis | no (target) | <code>bpred_survey_averaged.json</code>      |
| Required-vs-observed void shape (floor) | SHAPE-UNAVAILABLE (obs decline $\times 1.16$ vs. required $\times 3.31$ ; floor theorem)                                                                                                                                                                                                                                  | YES         | <code>R2_final.json</code>                   |
| WP-B dynamical three-phase              | CLOSED: TRACKS degenerately but $k = 0$ geometry misses BIC bar by 4953; T1 ceiling misses by 107 ( $\Delta\text{BIC} = +136$ )                                                                                                                                                                                           | YES         | <code>threephase_forced_geometry.json</code> |
| Buchert integrability                   | Not a single-phase GR solution: $\alpha^2$ drifts $\sim 152\%$ (forced, interior) / $\sim 5\times$ (observed, span) vs. tracker $\sim 2 \times 10^{-6}$                                                                                                                                                                   | YES         | <code>wpb_integrability.json</code>          |
| Void-wall contrast budget (two-sided)   | MARGINAL (field-mean 0.375–0.401 < required 0.474 at $z = 0$ ; deep-void+wall 0.46–0.57 reaches/exceeds it)                                                                                                                                                                                                               | partial     | <code>contrast_budget.json</code>            |
| Q-budget (Stage 0)                      | GATE=GO on empty/kinematic ceiling; physical budget 4–8 $\times$ short; required contrast exceeds empty-Milne ceiling $\times 1.36$                                                                                                                                                                                       | YES         | <code>q_budget.json</code>                   |
| Sound horizon / $r_d$                   | FAILS: in-model $r_d = 199.6$ Mpc (+36%); implied bare $\bar{H}_0$ 53.8 $\rightarrow \sim 40$ (fixed point 32.4)                                                                                                                                                                                                          | YES         | <code>wpc_sound_horizon.json</code>          |
| Growth / $S_8$                          | UNDEFINED-IN-KINEMATIC-READING (no perturbed metric $\Rightarrow$ no growth equation; measured $\sigma_m(z)$ is an input, not a prediction)                                                                                                                                                                               | YES         | <code>NOTES_growth.md</code>                 |
| $b_{\text{pred}}$ $z$ -profile (P4)     | predicts local $H_0$ bump decaying by $r_{\text{hom}} \sim 100 h^{-1}\text{Mpc}$ ; $r_{\text{hom}}^* = 140$ to reach $b_{\text{req}}$ is outside declared [80, 120] and beyond 2M++ homogeneity $\sim 137.5$                                                                                                              | partial     | <code>bpred_survey_averaged.json</code>      |

contrast reaching the required band within measured densities would *withdraw* the tension. The self-consistent field-averaged two-sided contrast is 0.375–0.401, below the required 0.474, but a deep-void-plus-overdense-wall combination reaches 0.46–0.57; neither refuted nor withdrawn, the pre-registered **Verdict: MARGINAL** (two-sided). The *Q-budget* Stage-0 pre-gate is an order-of-magnitude go/no-go on the Buchert backreaction  $Q$ , registered as NO-GO only if the most-permissive  $Q$ -ceiling falls below  $Q_{\text{req}}/3$  at every  $0.3 \leq z \leq 1.0$ : the gate is **GO** on the empty/kinematic ceiling, but the physical, lensing-capped budget runs 4–8 $\times$  short of  $Q_{\text{req}}$ , and the required contrast  $(H_v - H_w)/\bar{H}_0 = 0.516$  at  $z = 0$  exceeds the empty-Milne ceiling 0.379 by  $\times 1.36$ —unattainable by any physical void. Both rows corroborate the availability and dynamical closures above rather than carrying a verdict of their own.

## V. DISCUSSION

The result is *necessary but not sufficient*, made concrete. Paper I’s diagnosis was correct: the SN-versus-BAO+CMB split is the tracker’s one-parameter rigidity, and freeing  $f_v(z)$  removes it, reconciling the distance geometry below  $\Lambda\text{CDM}$  and collapsing the amplitude split from  $6.5\sigma$  to  $0.16\sigma$  (Sec. III). The reconciliation is nonetheless purely kinematic, and the gap between kinematics and dynamics is the whole story. A distance fit constrains only the redshift dependence of the expansion—the *shape* of  $f_v(z)$ —and shape freedom is cheap: a few extra spline nodes can bend a smooth  $E(z)$  onto the Hubble diagram. What a distance fit cannot see is whether that shape is realised as a void population in the sky, whether it solves the Buchert equations a real backreaction history must satisfy, whether it biases the local ladder by the amount the tension needs, and whether it calibrates to the early Universe. Each Part II test interrogates one of these, and each is a facet of the single fact that a kinematic target carries no dynamical or observational content beyond the distances it was fit-



ted to. The history is not present in the observed void population (Sec. IV A). It is not a single-phase GR solution, and the dynamical three-phase attempt to build one collapses back to the tracker (Sec. IV C). Its in-model sound horizon is too large to calibrate to the early Universe (Sec. IV D). And it predicts a local expansion-rate bias an order of magnitude too small (Sec. IV B).

A *positive* result would have required, jointly and specifically: an observed void population steep enough to supply the  $\times 3.31$  decline (not the observed  $\times 1.16$ ); a required history that is a single-phase integrability-consistent Buchert solution (not one drifting  $\sim 152\%$  in  $\alpha^2$ ); and an in-model sound horizon near 147 Mpc (not 199.6). None of these holds, and the local-bias prediction independently falls an order of magnitude short. The distinction between resolution and recasting from Paper I recurs sharply here: even where the model can be pushed toward a lower absolute Hubble rate, that is a recast of the tension—both canonical values reinterpreted as biased relative to a lower global average—not a reconciliation near the local  $H_0$ .

The  $S_8$ /growth question is out of reach of the kinematic reading altogether: it has no perturbed metric and hence no growth equation, so it cannot pose the question, let alone answer it (Sec. IV E). We flag this as the boundary of the kinematic reading, not as a computed defeat; a dynamical reading that could pose it is exactly the (closed) three-phase construction of Sec. IV C.

## VI. CONCLUSIONS

Free-history timescape does not resolve the Hubble tension. Freeing the void-fraction history from the one-parameter tracker removes the rigidity that Paper I identified and reconciles the SN+BAO+CMB distance geometry at joint  $\chi^2 = 1396.1$ , below flat  $\Lambda$ CDM (1402.2) and far below the tracker (1469.3), at a physically plausible  $f_{v0} = 0.640$ , with the amplitude split dissolving to  $0.16\sigma$ —though this last is a post-fit single-amplitude check with the jointly-fit shape held fixed, not an independent proof of data consistency. Even this geometric reconciliation is covariance-dependent: under the cosmology-independent supernova covariance of Refs. [17, 18] the fixed free-history is disfavoured relative to  $\Lambda$ CDM at every redshift cut. But the flexibility is kinematic, and it closes at every deeper physical test. The required history is not available: its  $\times 3.31$  decline is absent from the observed void population, which declines only  $\times 1.16$  (a floor theorem). The predicted local bias is  $+2.4\%$  against the  $+8.4\%$  required, the full admissi-

ble band below the target—a pre-registered PARTIAL that fails outright when the observed catalog is forced in. The dynamical three-phase construction meant to render the history a GR solution collapses back to the tracker, missing the geometry bar by 4953. And the model’s own sound horizon,  $r_d = 199.6$  Mpc ( $+36\%$ ), drives the bare rate to  $\sim 32\text{--}40 \text{ km s}^{-1} \text{ Mpc}^{-1}$ . Reconciliation without resolution. Wiltshire’s recent proposal that void backreaction supplies the cosmological constant and removes the need for a separate dark-energy source [31] is a qualitative, mechanism-level expectation; tested quantitatively and end to end, the freed mechanism reconciles the distance geometry but does not close the Hubble tension. This corroborates and sharpens Paper I: on present public data, eliminating dark energy through inhomogeneity—whether through the rigid tracker or its fully freed history—does not resolve the Hubble tension.

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## DATA AND CODE AVAILABILITY

Every quantitative claim reproduces from a committed script under the discipline *one number*  $\rightarrow$  *one script*  $\rightarrow$  *one result file*, with pre-registered thresholds fixed before each verdict. The supernova data are the public Pantheon+ release [19, 20]; the BAO and CMB inputs are DESI DR2 [5] and the Planck 2018 acoustic scale [3]; the observed void population is from BOSS DR12 [21] and the 2M++ field [26]. The code is public at <https://github.com/szhygulin/free-history-timescape>: this repository holds the Part I geometry-reconciliation code at the root and the Part II tension-test code under `tensions/`, and the former `free-history-timescape-tensions` repository is archived read-only on GitHub as the per-commit history of record. The code is released under the MIT license and will be deposited in a public archive on publication.

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